02 • TECHNICAL REQUIREMENTS

# Technical Requirements

What the site is built with, where it runs, and what it depends on.

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Planning documents for the Sali app

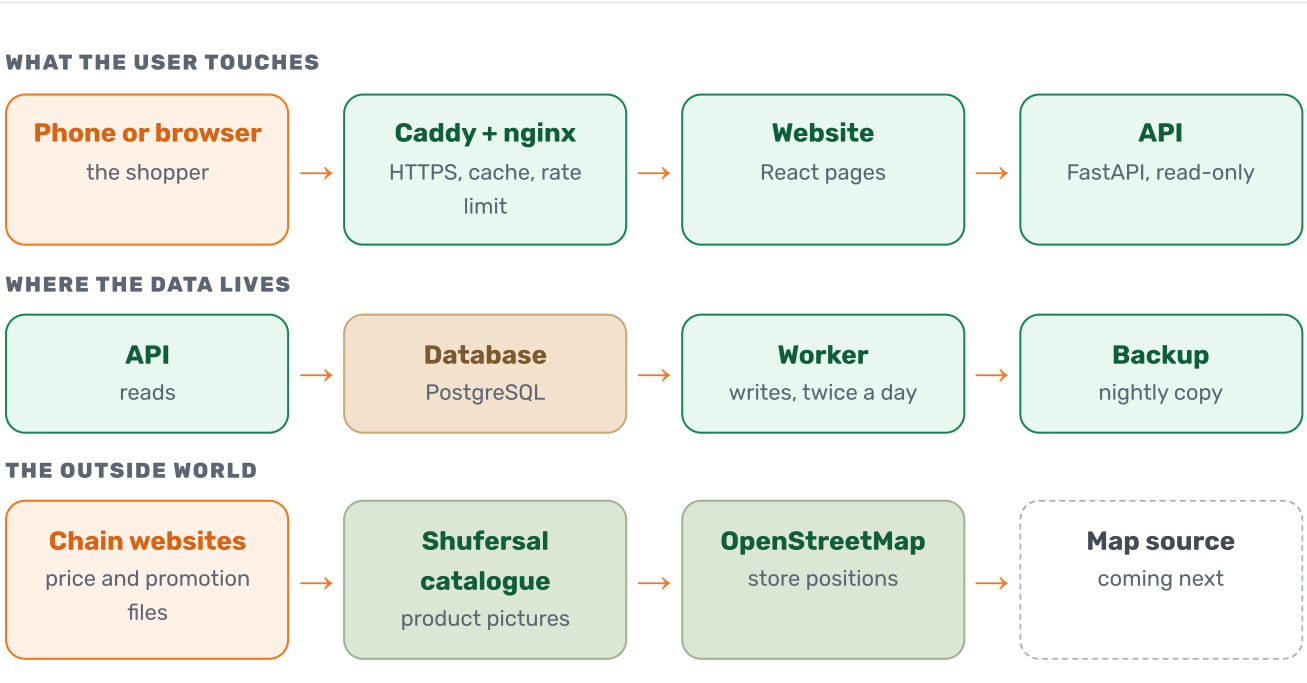
GitHub: supermarket-prices  Live site: sali.one

**IN ONE MINUTE**

- A **Worker** collects prices twice a day; a **website and API** show them. They meet in one **database**.
- Everything runs on **one server in Israel**, because 12 chains only answer Israeli addresses.
- The public side can only **read** the database, so it cannot change prices.

## 01 The architecture on one page

Solid boxes are live; the dashed box is coming next.



The Worker does all the slow work in the background, so no visitor waits for a download. The API only reads what is already stored.

## 02 What it is built with

| Part | Tool | In plain words |
|---|---|---|
| **Website** | React, TypeScript, Vite | Fast single-page site in Hebrew, right to left. Barcodes are read in the browser (ZXing). |
| **API** | Python 3.12, FastAPI | 14 endpoints: search, compare, basket, deals, cities, chains, pictures and more. |
| **Database** | PostgreSQL 16, pg_trgm | Open-source database; the add-on makes Hebrew search forgive spelling mistakes. |
| **Price collection** | Open-source scraper and parser | Community libraries that download and read every chain's files, pinned to a tested version. |
| **Running it** | Docker Compose, Caddy, nginx | Every part in its own box. Caddy gives sali.one its HTTPS certificate; nginx caches answers, limits 20 requests a second per visitor and adds security headers. |
| **Search visibility** | Server-built page headers | The server gives each page its own title, description and structured data, real 404s, and a sitemap that lists every priced product. |
| **Quality and release** | About 270 tests, GitHub Actions | Every change is tested against a temporary database; once it passes on main it is deployed to the server automatically. |

## 03 Where it runs, and what it depends on

| Part | Job | If it is down |
|---|---|---|
| **Caddy, nginx, website, API** | The only public side: serves sali.one over HTTPS and answers questions, read-only. | Site unavailable |
| **Worker** | At 06:00 and 18:00 Israel time: downloads and loads prices and promotions, refreshes the pre-worked tables, fills in pictures and positions (2 to 3 hours). | Data gets stale; an alert fires |
| **Database and backup** | All data; a nightly copy kept 7 days, also copied off the server. | Restore from backup |

| | | |
|---|---|---|
| **Chain websites** | The source of every price and promotion. | That chain stops updating; the rest carry on |
| **Shufersal catalogue, OpenStreetMap** | Product pictures; finding where each store is. | Placeholder picture; store left out of “near me” |
| **Waze, Google Maps, healthchecks.io** | Navigation links; alerts if the Worker or backup stops. | Link fails; no alert, site unaffected |

## 04 Safety and decisions already made

| Topic | What was decided, and why |
|---|---|
| **Read-only public side** | The API connects as a database user that can only read (and set a picture link). A flaw in the site cannot change prices. |
| **Limits** | Rate limits, caps on search length and basket size (100 items), and security headers. |
| **One server in Israel** | Instead of managed cloud services: size limits, network delay, and Israeli-only chains. |
| **Load 500 rows at a time** | One row at a time took 30 s where batches take 3.4 s for the same files. |
| **Full price files only** | A full file corrects itself; a changes-only file can miss something. |
| **Monitoring** | A health address fails if data is older than 36 hours, or a single chain is silent for 5 days, and an alert is sent. |